EXPERIMENT NO: 19 **Date: _____**

IMPLEMENTATION OF TRIGGER

AIM

To implement **Triggers** in PL/SQL for automatically updating a summary table whenever records are inserted into or deleted from a table.

OBJECTIVES

1. Create the **EMPLOYEE** table.
2. Create the **SALARY** table to maintain the employee count and total salary.
3. Insert an initial record into the **SALARY** table.
4. Create an **AFTER INSERT** trigger to automatically update the summary table whenever a new employee is added.
5. Create an **AFTER DELETE** trigger to automatically update the summary table whenever an employee record is deleted.
6. Verify the automatic updates by inserting and deleting employee records.

PROGRAM

1. Create EMPLOYEE Table

```
CREATE TABLE EMPLOYEE(
    EMP_ID NUMBER(5) PRIMARY KEY,
    EMP_NAME VARCHAR2(30),
    SALARY NUMBER(8)
);
```

2. Create SALARY Table

```
CREATE TABLE SALARY(
    EMP_COUNT NUMBER(5),
    TOTAL_SALARY NUMBER(10)
);
```

3. Initialize the SALARY Table

```
INSERT INTO SALARY VALUES(0,0);

COMMIT;
```

4. Create AFTER INSERT Trigger

```
CREATE OR REPLACE TRIGGER EMP_INSERT
AFTER INSERT
ON EMPLOYEE
FOR EACH ROW
BEGIN
    UPDATE SALARY
    SET EMP_COUNT = EMP_COUNT + 1,
        TOTAL_SALARY = TOTAL_SALARY + :NEW.SALARY;
END;
/
```

5. Create AFTER DELETE Trigger

```
CREATE OR REPLACE TRIGGER EMP_DELETE
AFTER DELETE
ON EMPLOYEE
FOR EACH ROW
BEGIN
    UPDATE SALARY
    SET EMP_COUNT = EMP_COUNT - 1,
        TOTAL_SALARY = TOTAL_SALARY - :OLD.SALARY;
```

```
END;
/
```

6. Insert Employee Records

```
INSERT INTO EMPLOYEE VALUES(101,'Arun',35000);
INSERT INTO EMPLOYEE VALUES(102,'Anu',45000);
INSERT INTO EMPLOYEE VALUES(103,'Rahul',40000);

COMMIT;
```

7. Display Summary Table

```
SELECT * FROM SALARY;
```

8. Delete an Employee Record

```
DELETE FROM EMPLOYEE
WHERE EMP_ID = 103;

COMMIT;
```

9. Display Summary Table Again

```
SELECT * FROM SALARY;
```

OUTPUT

1. Table Creation & Trigger Creation

```
Table created.

Table created.

1 row inserted.

Commit complete.

Trigger created.

Trigger created.
```

2. After Inserting Employee Records

```
3 rows inserted.

Commit complete.
EMP_COUNT  TOTAL_SALARY
---------  ------------
3          120000
```

3. After Deleting Employee with EMP_ID = 103

```
1 row deleted.

Commit complete.
EMP_COUNT  TOTAL_SALARY
---------  ------------
2          80000
```

RESULT

The **EMP_INSERT** and **EMP_DELETE** triggers were created and executed successfully. The **SALARY** table was automatically updated whenever employee records were inserted into or deleted from the **EMPLOYEE** table.